

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**First Semester of M. Tech Mechanical Engineering (Thermal) Examination****November 2018****ME502 Advanced Thermodynamics & Combustion****Date: 22.11.2018, Thursday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below.**

- (i) A can of soft drink at room temperature is put into the refrigerator so that it will cool. [02]
Would you model the can of soft drink as a closed system or as an open system?
Explain.
- (ii) In what forms energy can cross the boundaries of a closed system? Explain sign [02]
convention for energy transfer in close system with necessary diagrams.
- (iii) Define critical pressure. How does the boiling process at supercritical pressures differ [03]
from the boiling process at subcritical pressures?

Q – 2.a Prove that Clausius and Kelvin plank statements are equivalent to each other. [04]**Q – 2.b Answer any two questions. [12]**

- (i) Derive the equation of exergy for cyclic process having infinite and finite heat source.
- (ii) At the inlet to a certain nozzle the enthalpy of fluid passing is 2800 kJ/kg and the velocity is 50 m/s. At the discharge end the enthalpy is 2600 kJ/kg. The nozzle is horizontal and there is negligible heat loss from it. (i) Find the velocity at exit of the nozzle. (ii) If the inlet area is 900 cm² and the specific volume at inlet is 0.187 m³/kg, find the mass flow rate. (iii) If the specific volume at the nozzle exit is 0.498 m³/kg, find the exit area of nozzle.
- (iii) A domestic food refrigerator maintains a temperature of – 12°C. The ambient air temperature is 35°C. If heat leaks into the freezer at the continuous rate of 2 kJ/s determine the least power necessary to pump this heat out continuously.

Q - 3 Answer any two questions. [12]

- (i) An iron cube at a temperature of 400°C is dropped into an insulated bath containing 10 kg water at 25°C. The water finally reaches a temperature of 50°C at steady state. Given that the specific heat of water is equal to 4186 J/kg K. Find the entropy changes

for the iron cube and the water. Is the process reversible? If so why?

- (ii) 1.2 m³ of air is heated reversibly at constant pressure from 300 K to 600 K, and is then cooled reversibly at constant volume back to initial temperature. If the initial pressure is 1 bar, calculate: (i) The net heat flow (ii) The overall change in entropy. Represent the processes on T-S plot. Take $C_v = 0.718$ kJ/kg K and $R = 0.287$ kJ/kg K.
- (iii) Explain the term Lost work. Derive the equation of lost work for cyclic process.

SECTION – II

Q - 4 Answer the question below.

- (i) What is adiabatic flame temperature? [02]
- (ii) Compare enthalpy of combustion with enthalpy of formation. [02]
- (iii) What is isothermal compressibility? Derive its equation for ideal gas. [03]

Q – 5.a Describe Dalton's law of partial pressure. [02]

Q – 5.b Derive any three Maxwell's Relation. [07]

OR

Q – 5.b Derive Mayer's equation. [07]

Q – 5.c Determine the enthalpy of combustion of liquid octane (C₈H₁₈) at 25°C and 1 atm, using [05]
following data of enthalpy of formation at 25°C and 1 atm.

-393,520 kJ/K mol for CO₂,

-285,830 kJ/K mol for H₂O(liquid),

and -249,950 kJ/K mol for C₈H₁₈(Liquid)

Assume the water in the products is in the liquid form.

Q – 6.a Which of the instrument is used to analyze products of combustion? Explain [07]
construction and working with neat sketch.

Q – 6.b For an adiabatic mixing of two fluids having same masses and specific heat but at [07]
different temperatures, prove that entropy change of the universe is given by –

$$\Delta S_{univ} = 2mc \ln \frac{(T_1 + T_2)/2}{\sqrt{T_1 T_2}}$$

OR

Q – 6.a Derive an energy balance relation for a reacting steady flow open system. [07]

Q – 6.b An insulated rigid tank is divided into two compartments by a partition. One [07]
compartment contains 7 kg of oxygen gas at 40°C and 100 kPa, and the other

compartment contains 4 kg of nitrogen gas at 20°C and 150 kPa. Now the partition is removed, and the two gases are allowed to mix. Determine (a) the mixture temperature and (b) the mixture pressure after equilibrium has been established.
